



Figure 9: The main loop of health-aware decision making

implement the proposed approach is left for future publications, however examples of DM algorithms that can deal with state or outcome uncertainty are provided by ? (?). In systems where both state and outcome uncertainty are not a factor, a variety of traditional state space planning algorithms (?) would not need any modification to produce plans for spaces of state vectors that include health-related components.

5.3 Emergency response vs. health management

For realistic complex systems operating in the presence of state and outcome uncertainty, S , B , and O are likely to be infinitely large (with $|A| \ll \infty$, although still potentially large). The problem of finding exact optimal policies in such cases is PSPACE-complete (?). Approximate solution methods typically work *online*, constructing π for the current belief b based on beliefs reachable from b within the decision horizon (?). They also typically perform targeted sampling from S/B , A , and O , thus optimality guarantees can be harder to provide. We, therefore, propose the following:

- That **system emergency response (SER)** be defined as an automated or semi-automated process that is invoked to maximize the likelihood of preserving the system’s integrity, regardless of the effect on operational goals (*i.e.*, a different objective from HADM).
- That emergency response policy π_{SER} be computed separately from the HADM policy.

In the space domain, an example of SER is putting a spacecraft in a *safe mode* until the emergency is resolved (?). In aviation, it could be executing recommendations of a collision avoidance system (?). Introduction of a separate SER system would likely require the introduction of S_{SER} , an additional S subset which defines for which states π_{SER} is invoked versus π_{HADM} . Once again, however, S_{SER} will not necessarily only contain system fault and failure states. For instance, states where environmental conditions warrant emergency response (*e.g.*, solar activity interrupting teleoperation of the rover) would be included as well. Since the scope of the SER problem is likely to be much narrower than that of HADM, it opens up the possibility of computing, verifying, and validating π_{SER} *offline* (?).

As sensors, computing capabilities, and DM algorithms improve further, the fraction of the system’s state space that is under the purview of SER will decrease. Still, we foresee the need for a “safety net” SER to be there for safety-critical functions and thus advocate SER independence. SER would cover the situations where the primary HADM system may not be able to produce a suitable solution in a desired amount of time, serving as an equivalent of human reflexive responses triggered by important stimuli versus the more

deliberative cognitive functionality of the brain. It could also provide dissimilar redundancy for critical regions of S , essentially implementing the Swiss Cheese Model (?), which advocates multiple different layers of protection for important functions.

5.4 Remarks on computational complexity

A counter-argument to the unified approach can be made that it increases the computational complexity by operating on larger S/B , A , and O spaces. While this is indeed a concern, algorithms have been developed in the recent years that can effectively accommodate much of the complexity, even for problems with state and outcome uncertainty, by approximating optimal policies online (??; ?). Secondly, some of the complexity is removed from HADM into an independent SER component, making the former problem easier. Finally, as noted above, unifying the SHM and DM elements may also result in reduction of overlapping model variables.

6 Conclusions

This paper reexamines the prevalent approach to performing system health management and makes the case for why keeping system health management functions separate from decision making functions can be inefficient and/or ineffective. We also present the case for why prognostics is only meaningful in a limited set of circumstances and, even then, needs to be driven by decision making requirements.

We then explain the rationale for unifying (not just integrating) system health management with decision making and outline an approach for accomplishing that. We also propose keeping emergency response functionality separate to guarantee timely response, provide dissimilar redundancy, and allow for offline computation, validation, and verification of emergency response policies.

We believe that the proposed unification approach will improve the performance of both decision making and system health management, while potentially simplifying informational architectures, reducing sensor suite sizes, and combining modeling efforts. The approach is scalable with respect to system complexity and the types of uncertainties present. It also puts a variety of existing and emerging computational methods at the disposal of system designers.

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